

MOORED BUOY 22001, Japan

WMO: 992300

Lat: 28.300N

Lon: 126.000E

Elev: 0

StdP: 101.33

Time Zone: 8.00 (E08)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.4	10.3	1.7	4.3	10.7	2.5	4.5	11.2	17.3	13.3	16.2	13.0	11.2	0	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.4	30.2	27.1	30.0	27.1	29.7	26.9	28.1	29.5	27.7	29.2	27.5	29.1	6.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.7	23.6	29.2	27.2	23.0	29.0	27.0	22.7	28.8	90.2	29.6	88.7	29.4	87.3	29.2	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	15.8	13.7	8.0	30.7	1.4	0.5	7.0	31.0	6.2	31.3	5.4	31.6	4.4	32.0		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.8	15.2	14.6	16.7	19.3	22.4	25.6	28.4	28.5	26.9	24.1	21.3	17.6	(6)
(7)		DBStd	5.32	2.76	3.07	2.88	2.82	2.08	1.81	1.00	0.88	1.42	1.81	2.30	2.70	(7)
(8)		HDD10.0	2	1	1	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	356	103	111	68	22	2	0	0	0	0	0	4	47	(9)
(10)		CDD10.0	4294	161	129	207	281	386	469	570	573	508	438	338	235	(10)
(11)		CDD18.3	1608	6	6	16	52	129	219	312	314	258	180	92	24	(11)
(12)		CDH23.3	13661	0	0	3	67	379	1752	3781	3883	2651	952	188	6	(12)
(13)	CDH26.7	3638	0	0	0	0	6	252	1337	1417	589	37	1	0	(13)	
(14)	Wind	WSAvg	7.4	8.4	8.2	7.6	6.7	6.0	6.4	6.9	7.4	7.7	7.9	7.6	7.7	(14)
(15)	Precipitation	PrecAvg	1936	73	86	136	188	227	344	131	197	243	129	107	75	(15)
(16)		PrecMax	2577	155	242	368	412	392	523	440	462	482	286	262	173	(16)
(17)		PrecMin	1376	53	48	35	92	71	86	12	15	29	39	52	45	(17)
(18)		PrecStd	310	28	53	66	63	84	108	107	118	113	66	48	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.5	22.6	23.8	25.5	27.4	29.2	30.6	30.7	30.0	28.2	26.7	24.0	(19)
(20)			MCWB	20.4	20.8	22.3	23.9	25.7	27.0	27.2	26.9	27.0	25.6	22.4	21.5	(20)
(21)		2%	DB	21.5	21.7	22.8	24.7	26.3	28.7	30.2	30.2	29.5	27.6	25.9	23.2	(21)
(22)			MCWB	18.8	19.6	21.6	23.2	24.8	26.9	27.2	27.0	26.7	25.1	23.1	20.5	(22)
(23)		5%	DB	20.5	20.7	22.1	24.2	25.7	28.3	29.9	30.0	29.2	27.1	25.1	22.4	(23)
(24)			MCWB	17.5	18.5	21.0	22.9	23.9	26.8	27.2	27.0	26.3	24.6	22.5	19.5	(24)
(25)		10%	DB	19.4	19.4	21.2	23.4	25.1	28.0	29.7	29.7	28.8	26.6	24.4	21.6	(25)
(26)			MCWB	16.7	17.2	19.7	22.0	23.2	26.7	27.0	26.8	26.0	23.8	21.6	18.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	21.0	22.6	24.5	26.0	28.0	28.7	28.2	27.7	25.8	23.9	22.8	(27)
(28)			MADB	22.3	21.7	23.3	25.3	26.5	28.7	29.8	29.6	29.3	27.4	25.1	23.6	(28)
(29)		2%	WB	19.9	19.4	21.8	23.9	25.2	27.5	28.3	27.7	27.4	25.4	23.2	21.2	(29)
(30)			MADB	21.2	20.5	22.6	24.6	25.9	28.3	29.6	29.5	29.1	27.1	24.7	23.0	(30)
(31)		5%	WB	18.5	18.2	20.8	23.4	24.6	27.2	28.0	27.5	27.0	24.5	22.6	19.8	(31)
(32)			MADB	20.1	19.8	21.7	24.2	25.4	28.1	29.5	29.3	28.9	26.2	24.2	21.8	(32)
(33)		10%	WB	17.3	16.9	19.3	22.8	24.0	27.0	27.6	27.2	26.5	23.4	21.9	18.6	(33)
(34)			MADB	19.3	18.8	20.6	23.7	24.9	27.9	29.3	29.1	28.5	25.4	23.7	20.9	(34)
(35)	Mean Daily Temperature Range	MDBR	2.9	3.0	3.5	3.3	2.6	2.1	1.4	1.4	1.5	1.7	2.3	2.4	(35)	
(36)		5% DB	MADB	4.5	4.1	4.1	3.4	2.9	1.8	1.3	1.4	1.6	1.7	2.5	3.2	(36)
(37)			MCWBR	4.8	4.7	5.0	3.9	2.7	1.1	1.2	1.2	1.6	1.9	2.6	3.6	(37)
(38)		5% WB	MADB	4.5	4.3	4.9	3.6	2.9	1.5	1.4	1.4	1.4	1.8	3.2	3.2	(38)
(39)			MCWBR	5.4	5.2	5.5	4.2	3.0	1.3	1.4	1.5	1.5	2.0	3.3	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.490	0.532	0.591	0.599	0.551	0.531	0.502	0.508	0.504	0.494	0.485	0.481	(40)	
(41)		taud	1.982	1.905	1.793	1.834	2.018	2.138	2.238	2.217	2.186	2.128	2.085	2.012	(41)	
(42)		Ebn at Noon	725	733	717	730	766	777	799	792	783	761	728	712	(42)	
(43)		Edh at Noon	155	181	214	212	177	156	141	143	143	143	139	144	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.98	2.48	3.42	4.14	4.71	4.68	6.51	5.95	4.95	3.90	2.64	2.05	(44)	
(45)		RadStd	0.36	0.50	0.37	0.43	0.62	0.49	0.63	0.39	0.44	0.32	0.26	0.24	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page